

Databricks

Exam Questions Databricks-Certified-Data-Analyst-Associate

Databricks Certified Data Analyst Associate Exam



NEW QUESTION 1

A data organization has a team of engineers developing data pipelines following the medallion architecture using Delta Live Tables. While the data analysis team working on a project is using gold-layer tables from these pipelines, they need to perform some additional processing of these tables prior to performing their analysis.

Which of the following terms is used to describe this type of work?

- A. Data blending
- B. Last-mile
- C. Data testing
- D. Last-mile ETL
- E. Data enhancement

Answer: D

Explanation:

Last-mile ETL is the term used to describe the additional processing of data that is done by data analysts or data scientists after the data has been ingested, transformed, and stored in the lakehouse by data engineers. Last-mile ETL typically involves tasks such as data cleansing, data enrichment, data aggregation, data filtering, or data sampling that are specific to the analysis or machine learning use case. Last-mile ETL can be done using Databricks SQL, Databricks notebooks, or Databricks Machine Learning. References: Databricks - Last-mile ETL, Databricks - Data Analysis with Databricks SQL

NEW QUESTION 2

Consider the following two statements:

Statement 1:

```
SELECT *
FROM customers
LEFT SEMI JOIN orders
ON customers.customer_id = orders.customer_id;
```

Statement 2:

```
SELECT *
FROM customers
LEFT ANTI JOIN orders
ON customers.customer_id = orders.customer_id;
```

Which of the following describes how the result sets will differ for each statement when they are run in Databricks SQL?

- A. The first statement will return all data from the customers table and matching data from the orders table
- B. The second statement will return all data from the orders table and matching data from the customers table
- C. Any missing data will be filled in with NULL.
- D. When the first statement is run, only rows from the customers table that have at least one match with the orders table on customer_id will be returned
- E. When the second statement is run, only those rows in the customers table that do not have at least one match with the orders table on customer_id will be returned.
- F. There is no difference between the result sets for both statements.
- G. Both statements will fail because Databricks SQL does not support those join types.
- H. When the first statement is run, all rows from the customers table will be returned and only the customer_id from the orders table will be returned
- I. When the second statement is run, only those rows in the customers table that do not have at least one match with the orders table on customer_id will be returned.

Answer: B

Explanation:

Based on the images you sent, the two statements are SQL queries for different types of joins between the customers and orders tables. A join is a way of combining the rows from two table references based on some criteria. The join type determines how the rows are matched and what kind of result set is returned. The first statement is a query for a LEFT SEMI JOIN, which returns only the rows from the left table reference (customers) that have a match with the right table reference (orders) on the join condition (customer_id). The second statement is a query for a LEFT ANTI JOIN, which returns only the rows from the left table reference (customers) that have no match with the right table reference (orders) on the join condition (customer_id). Therefore, the result sets for the two statements will differ in the following way:

? The first statement will return a subset of the customers table that contains only the customers who have placed at least one order. The number of rows returned will be less than or equal to the number of rows in the customers table, depending on how many customers have orders. The number of columns returned will be the same as the number of columns in the customers table, as the LEFT SEMI JOIN does not include any columns from the orders table.

? The second statement will return a subset of the customers table that contains only the customers who have not placed any order. The number of rows returned will be less than or equal to the number of rows in the customers table, depending on how many customers have no orders. The number of columns returned will be the same as the number of columns in the customers table, as the LEFT ANTI

JOIN does not include any columns from the orders table. The other options are not correct because:

? A. The first statement will not return all data from the customers table, as it will exclude the customers who have no orders. The second statement will not return all data from the orders table, as it will exclude the orders that have a matching

customer. Neither statement will fill in any missing data with NULL, as they do not return any columns from the other table.

? C. There is a difference between the result sets for both statements, as explained

above. The LEFT SEMI JOIN and the LEFT ANTI JOIN are not equivalent operations and will produce different outputs.

? D. Both statements will not fail, as Databricks SQL does support those join types.

Databricks SQL supports various join types, including INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, LEFT SEMI, LEFT ANTI, and CROSS. You can also use NATURAL, USING, or LATERAL keywords to specify different join criteria.

? E. The first statement will not return only the customer_id from the orders table, as

it will return all columns from the customers table. The second statement is correct, but it is not the only difference between the result sets.

References: JOIN | Databricks on AWS, JOIN - Azure Databricks - Databricks SQL | Microsoft Learn, array_join function | Databricks on AWS, Hints | Databricks on AWS

NEW QUESTION 3

An analyst writes a query that contains a query parameter. They then add an area chart visualization to the query. While adding the area chart visualization to a dashboard, the analyst chooses "Dashboard Parameter" for the query parameter associated with the area chart.

Which of the following statements is true?

A. The area chart will use whatever is selected in the Dashboard Parameter while all or the other visualizations will remain changed regardless of their parameter use.

B. The area chart will use whatever is selected in the Dashboard Parameter along with all of the other visualizations in the dashboard that use the same parameter.

C. The area chart will use whatever value is chosen on the dashboard at the time the area chart is added to the dashboard.

D. The area chart will use whatever value is input by the analyst when the visualization is added to the dashboar

E. The parameter cannot be changed by the user afterwards.

F. The area chart will convert to a Dashboard Parameter.

Answer: B

Explanation:

A Dashboard Parameter is a parameter that is configured for one or more visualizations within a dashboard and appears at the top of the dashboard. The parameter values specified for a Dashboard Parameter apply to all visualizations reusing that particular Dashboard Parameter¹. Therefore, if the analyst chooses ??Dashboard Parameter?? for the query parameter associated with the area chart, the area chart will use whatever is selected in the Dashboard Parameter along with all of the other visualizations

in the dashboard that use the same parameter. This allows the user to filter the data across multiple visualizations using a single parameter widget². References: Databricks SQL dashboards, Query parameters

NEW QUESTION 4

A data analyst has recently joined a new team that uses Databricks SQL, but the analyst has never used Databricks before. The analyst wants to know where in Databricks SQL they can write and execute SQL queries.

On which of the following pages can the analyst write and execute SQL queries?

A. Data page

B. Dashboards page

C. Queries page

D. Alerts page

E. SQL Editor page

Answer: E

Explanation:

The SQL Editor page is where the analyst can write and execute SQL queries in Databricks SQL. The SQL Editor page has a query pane where the analyst can type or paste SQL statements, and a results pane where the analyst can view the query results in a table or a chart. The analyst can also browse data objects, edit multiple queries, execute a single query or multiple queries, terminate a query, save a query, download a query result, and more from the SQL Editor page.

References: Create a query in SQL editor

NEW QUESTION 5

A data analyst needs to use the Databricks Lakehouse Platform to quickly create SQL queries and data visualizations. It is a requirement that the compute resources in the platform can be made serverless, and it is expected that data visualizations can be placed within a dashboard.

Which of the following Databricks Lakehouse Platform services/capabilities meets all of these requirements?

A. Delta Lake

B. Databricks Notebooks

C. Tableau

D. Databricks Machine Learning

E. Databricks SQL

Answer: E

Explanation:

Databricks SQL is a serverless data warehouse on the Lakehouse that lets you run all of your SQL and BI applications at scale with your tools of choice, all at a fraction of the cost of traditional cloud data warehouses¹. Databricks SQL allows you to create SQL queries and data visualizations using the SQL Analytics UI or the Databricks

SQL CLI². You can also place your data visualizations within a dashboard and share it with other users in your organization³. Databricks SQL is powered by Delta Lake, which provides reliability, performance, and governance for your data lake⁴. References:

? Databricks SQL

? Query data using SQL Analytics

? Visualizations in Databricks notebooks

? Delta Lake

NEW QUESTION 6

After running `DESCRIBE EXTENDED accounts.customers;`, the following was returned:

Name	accounts.customers
Location	dbfs:/stakeholders/customers
Provider	delta
Owner	root
Type	EXTERNAL

Now, a data analyst runs the following command:

`DROP accounts.customers;`

Which of the following describes the result of running this command?

- A. Running `SELECT * FROM delt`
- B. ``dbfs:/stakeholders/customers`` results in an error.
- C. Running `SELECT * FROM accounts.customers` will return all rows in the table.
- D. All files with the `.customers` extension are deleted.
- E. The `accounts.customers` table is removed from the metastore, and the underlying data files are deleted.
- F. The `accounts.customers` table is removed from the metastore, but the underlying data files are untouched.

Answer: E

Explanation:

the `accounts.customers` table is an `EXTERNAL` table, which means that it is stored outside the default warehouse directory and is not managed by Databricks. Therefore, when you run the `DROP` command on this table, it only removes the metadata information from the metastore, but does not delete the actual data files from the file system. This means that you can still access the data using the location path (`dbfs:/stakeholders/customers`) or create another table pointing to the same location. However, if you try to query the table using its name (`accounts.customers`), you will get an error because the table no longer exists in the metastore. References: `DROP TABLE` | Databricks on AWS, Best practices for dropping a managed Delta Lake table - Databricks

NEW QUESTION 7

Delta Lake stores table data as a series of data files, but it also stores a lot of other information.

Which of the following is stored alongside data files when using Delta Lake?

- A. None of these
- B. Table metadata, data summary visualizations, and owner account information
- C. Table metadata
- D. Data summary visualizations
- E. Owner account information

Answer: C

Explanation:

Delta Lake stores table data as a series of data files in a specified location, but it also stores table metadata in a transaction log. The table metadata includes the schema, partitioning information, table properties, and other configuration details. The table metadata is stored alongside the data files and is updated atomically with every write operation. The table metadata can be accessed using the `DESCRIBE DETAIL` command or the `DeltaTable` class in Scala, Python, or Java. The table metadata can also be enriched with custom tags or user-defined commit messages using the `TBLPROPERTIES` or `userMetadata` options. References:

? Enrich Delta Lake tables with custom metadata

? Delta Lake Table metadata - Stack Overflow

? Metadata - The Internals of Delta Lake

NEW QUESTION 8

Which of the following statements about adding visual appeal to visualizations in the Visualization Editor is incorrect?

- A. Visualization scale can be changed.
- B. Data Labels can be formatted.
- C. Colors can be changed.
- D. Borders can be added.
- E. Tooltips can be formatted.

Answer: D

Explanation:

The Visualization Editor in Databricks SQL allows users to create and customize various types of charts and visualizations from the query results. Users can change the visualization type, select the data fields, adjust the colors, format the data labels, and modify the tooltips. However, there is no option to add borders to the visualizations in the Visualization Editor. Borders are not a supported feature of the new chart visualizations in Databricks¹. Therefore, the statement that borders can be added is incorrect. References:

? New chart visualizations in Databricks | Databricks on AWS

NEW QUESTION 9

A data analysis team is working with the `table_bronze` SQL table as a source for one of its most complex projects. A stakeholder of the project notices that some of the downstream data is duplicative. The analysis team identifies `table_bronze` as the source of the duplication.

Which of the following queries can be used to deduplicate the data from table_bronze and write it to a new table table_silver?

- A)
CREATE TABLE table_silver AS SELECT DISTINCT *
FROM table_bronze;
- B)
CREATE TABLE table_silver AS INSERT *
FROM table_bronze;
- C)
CREATE TABLE table_silver AS MERGE DEDUPLICATE *
FROM table_bronze;
- D)
INSERT INTO TABLE table_silver SELECT * FROM table_bronze;
- E)
INSERT OVERWRITE TABLE table_silver SELECT * FROM table_bronze;

- A. Option A
B. Option B
C. Option C
D. Option D
E. Option E

Answer: A

Explanation:

Option A uses the SELECT DISTINCT statement to remove duplicate rows from the table_bronze and create a new table table_silver with the deduplicated data. This is the correct way to deduplicate data using Spark SQL¹². Option B simply inserts all the rows from table_bronze into table_silver, without removing any duplicates. Option C is not a valid syntax for Spark SQL, as there is no MERGE DEDUPLICATE statement. Option D appends all the rows from table_bronze into table_silver, without removing any duplicates. Option E overwrites the existing data in table_silver with the data from table_bronze, without removing any duplicates. References: Delete Duplicate using SPARK SQL, Spark SQL - How to Remove Duplicate Rows

NEW QUESTION 10

A data analyst has set up a SQL query to run every four hours on a SQL endpoint, but the SQL endpoint is taking too long to start up with each run. Which of the following changes can the data analyst make to reduce the start-up time for the endpoint while managing costs?

- A. Reduce the SQL endpoint cluster size
B. Increase the SQL endpoint cluster size
C. Turn off the Auto stop feature
D. Increase the minimum scaling value
E. Use a Serverless SQL endpoint

Answer: E

Explanation:

A Serverless SQL endpoint is a type of SQL endpoint that does not require a dedicated cluster to run queries. Instead, it uses a shared pool of resources that can scale up and down automatically based on the demand. This means that a Serverless SQL endpoint can start up much faster than a SQL endpoint that uses a cluster, and it can also save costs by only paying for the resources that are used. A Serverless SQL endpoint is suitable for ad-hoc queries and exploratory analysis, but it may not offer the same level of performance and isolation as a SQL endpoint that uses a cluster. Therefore, a data analyst should consider the trade-offs between speed, cost, and quality when choosing between a Serverless SQL endpoint and a SQL endpoint that uses a cluster. References: Databricks SQL endpoints, Serverless SQL endpoints, SQL endpoint clusters

NEW QUESTION 10

Which of the following statements describes descriptive statistics?

- A. A branch of statistics that uses summary statistics to quantitatively describe and summarize data.
B. A branch of statistics that uses a variety of data analysis techniques to infer properties of an underlying distribution of probability.
C. A branch of statistics that uses quantitative variables that must take on a finite or countably infinite set of values.
D. A branch of statistics that uses summary statistics to categorically describe and summarize data.
E. A branch of statistics that uses quantitative variables that must take on an uncountable set of values.

Answer: A

Explanation:

Descriptive statistics is a branch of statistics that uses summary statistics, such as mean, median, mode, standard deviation, range, frequency, or correlation, to quantitatively describe and summarize data. Descriptive statistics can help data analysts understand the main features of a data set, such as its central tendency, variability, or distribution. Descriptive statistics can also help data analysts visualize data using charts, graphs, or tables. Descriptive statistics do not make any inferences or predictions about the data, unlike inferential statistics, which use data analysis techniques to infer properties of an underlying population or probability distribution from a sample of data. References: Databricks - Descriptive Statistics, Databricks - Data Analysis with Databricks SQL

NEW QUESTION 13

A data analyst runs the following command: SELECT age, country
FROM my_table
WHERE age >= 75 AND country = 'canada';
Which of the following tables represents the output of the above command?

- A)

age	country
80	canada
<i>NULL</i>	canada
90	<i>NULL</i>

B)

age	country
80	<i>NULL</i>
75	<i>NULL</i>
90	<i>NULL</i>

C)

id	age	country
900	80	canada
901	75	canada
902	90	canada

D)

age	country
80	canada
14	canada
90	canada

E)

age	country
80	canada
75	canada
90	canada

- A. Option A
- B. Option B
- C. Option C
- D. Option D
- E. Option E

Answer: E

Explanation:

The SQL query provided is designed to filter out records from `my_table` where the age is 75 or above and the country is Canada. Since I can't view the content of the links provided directly, I need to rely on the image attached to this question for context. Based on that, Option E (the image attached) represents a table with columns `age` and `country`, showing records where age is 75 or above and country is Canada. References: The answer can be inferred from understanding SQL queries and their outputs as per Databricks documentation: Databricks SQL

NEW QUESTION 17

A data analyst is attempting to drop a table `my_table`. The analyst wants to delete all table metadata and data. They run the following command: `DROP TABLE IF EXISTS my_table;` While the object no longer appears when they run `SHOW TABLES`, the data files still exist. Which of the following describes why the data files still exist and the metadata files were deleted?

- A. The table's data was larger than 10 GB
- B. The table did not have a location
- C. The table was external
- D. The table's data was smaller than 10 GB
- E. The table was managed

Answer: C

Explanation:

An external table is a table that is defined in the metastore, but its data is stored outside of the Databricks environment, such as in S3, ADLS, or GCS. When an external table is dropped, only the metadata is deleted from the metastore, but the data files are not affected. This is different from a managed table, which is a table whose data is stored in the Databricks environment, and whose data files are deleted when the table is dropped. To delete the data files of an external table, the analyst needs to specify the `PURGE` option in the `DROP TABLE` command, or manually delete the files from the storage system. References: `DROP TABLE`, Drop Delta table features, Best practices for dropping a managed Delta Lake table

NEW QUESTION 19

Which of the following describes how Databricks SQL should be used in relation to other business intelligence (BI) tools like Tableau, Power BI, and Looker?

- A. As an exact substitute with the same level of functionality
- B. As a substitute with less functionality
- C. As a complete replacement with additional functionality
- D. As a complementary tool for professional-grade presentations
- E. As a complementary tool for quick in-platform BI work

Answer: E

Explanation:

Databricks SQL is not meant to replace or substitute other BI tools, but rather to complement them by providing a fast and easy way to query, explore, and visualize data on the lakehouse using the built-in SQL editor, visualizations, and dashboards. Databricks SQL also integrates seamlessly with popular BI tools like Tableau, Power BI, and Looker, allowing analysts to use their preferred tools to access data through Databricks clusters and SQL warehouses. Databricks SQL offers low-code and no-code experiences, as well as optimized connectors and serverless compute, to enhance the productivity and performance of BI workloads on the lakehouse. References: Databricks SQL, Connecting Applications and BI Tools to Databricks SQL, Databricks integrations overview, Databricks SQL: Delivering a Production SQL Development Experience on the Lakehouse

NEW QUESTION 23

Which of the following is an advantage of using a Delta Lake-based data lakehouse over common data lake solutions?

- A. ACID transactions
- B. Flexible schemas
- C. Data deletion
- D. Scalable storage
- E. Open-source formats

Answer: A

Explanation:

A Delta Lake-based data lakehouse is a data platform architecture that combines the scalability and flexibility of a data lake with the reliability and performance of a data warehouse. One of the key advantages of using a Delta Lake-based data lakehouse over common data lake solutions is that it supports ACID transactions, which ensure data integrity and consistency. ACID transactions enable concurrent reads and writes, schema enforcement and evolution, data versioning and rollback, and data quality checks. These features are not available in traditional data lakes, which rely on file-based storage systems that do not support transactions. References:

? Delta Lake: Lakehouse, warehouse, advantages | Definition

? Synapse – Data Lake vs. Delta Lake vs. Data Lakehouse

? Data Lake vs. Delta Lake - A Detailed Comparison

? Building a Data Lakehouse with Delta Lake Architecture: A Comprehensive Guide

NEW QUESTION 27

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